Query: what support is there for the idea that two FM agents debating (or simply two FMs) leads to better answers?

Short answer

The wiki supports the narrow claim — two oppositely-prompted agents debating can correct a single model's confident error — but the support rests almost entirely on one source, Liang2024EncouragingDT (Multi-Agent Debate, EMNLP 2024). The broader claim ("simply two FMs" helps, or "more debate is better") is not supported and is partly contradicted by that same source.

What is supported: debate as a cure for a specific failure mode

The mechanism is grounded, not just asserted. Liang2024EncouragingDT names Degeneration-of-Thought (DoT): once a single model is confident, self-reflection stops producing new thoughts even when the first answer was wrong — "Once the LLM-based agent has established confidence in its answers, it is unable to generate novel thoughts later through self-reflection even if the initial stance is incorrect" (Liang2024EncouragingDT `sec. 1, p. 17889`).

Multi-Agent Debate is the proposed fix: an affirmative and a negative debater argue over a shared transcript under a judge, so "the distorted thinking of one agent can be corrected by the others" (Liang2024EncouragingDT `sec. 2, p. 17890`). The value is supplying the external perspective a lone model structurally lacks — see why to communicate.

The claim is demonstrated rather than rhetorical: across forced iterations, a self-reflecting model's between-round answer barely changes (it stops disagreeing with itself), while a two-agent debate keeps producing divergence (Liang2024EncouragingDT `sec. 1, p. 17889, fig. 1`). The benefit shows up on tasks where the instinctive first answer is systematically wrong — commonsense translation and counter-intuitive arithmetic (Liang2024EncouragingDT `sec. 3.2, p. 17892`).

Chen2026TheFW (the survey) gives this a general framing: a competitive setting is "not always purely adversarial — some systems use it, through debate or deliberation, to surface stronger arguments or reach more accurate group outcomes." This is a second source endorsing the idea in principle, but it is a survey-level generalization, not independent evidence.

What is NOT supported

- **"More agents / more debate is better" is contradicted.** Multi-Agent Debate's own disconfirming evidence: beyond the default pair, performance can drop because the backbones handle the longer transcript poorly, and forcing rounds past the judge's adaptive break degrades the result (Liang2024EncouragingDT `sec. 4.3, fig. 5, p. 17896`). The benefit is bounded by the model's long-context ability, not monotone in the amount of debate.
- **"Simply two FMs" (collaboration without engineered disagreement) is not addressed as a better-answer claim.** The wiki's debate evidence is specifically about *opposing-stance* agents plus a designed stopping rule and a tuned disagreement level ("tit for tat", four levels; modest is best, total disagreement over-polarizes — Liang2024EncouragingDT `app. D, tab. 11, p. 17902`). The wiki has no source showing that two undifferentiated models, or two cooperating models, beat one. DyLAN is multi-agent but explicitly "not a debate framework," and the wiki does not record a head-to-head "two beats one" result for it.

Why the support is fragile

- **Single-source.** Both Multi-Agent Debate and Degeneration-of-Thought are `source_count: 1`. *[tentative]*
- **Cure and diagnosis share authors and task selection.** The DoT evidence comes from the same authors who propose the fix, on tasks chosen because instinctive answers are wrong; how prevalent DoT is on tasks where first answers are usually right is not established (see Degeneration-of-Thought, Disconfirming Evidence).
- **Benefit entangled with a biased judge.** The judge "shows a preference to the side with the same LLM as the backbone" (Liang2024EncouragingDT `sec. 1, p. 17890`), and in the default setup the judge and debaters share a backbone — so how much of the gain is the divergent debate versus a well-timed adaptive stop is not cleanly separated (see Appraisal and Open Questions).

Bottom line

Supported: two oppositely-stanced FM agents, with a judge and a tuned disagreement level, can beat single-agent self-reflection on traps where the first answer is reliably wrong — on one well-grounded but single-author source, with a generalizing nod from the survey. Not supported: that adding agents/rounds keeps helping (contradicted), or that two undifferentiated/cooperating FMs beat one (the wiki is silent). For a high-stakes claim, this needs an independent replication of DoT and a framework that separates the debate's benefit from the judge's adaptive break — both flagged as open threads in hot.